

Title: Updatable Meta-Weights Architecture: A Unified Path from Probabilistic Correlation to General Intelligence

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ABSTRACT

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What is the fundamental difference between human intelligence and today's large language models? Mainstream research oscillates between scaling up data (emergentists) and injecting explicit rules (symbolists), yet neither captures the core mechanism that allows humans to transcend pure statistical association. We propose the Updatable Meta-Weights Architecture (UMWA), a unified theoretical framework that defines intelligence as the interplay between a Base Experience Network (probabilistic association) and a Meta-Rules Verification Layer (high-priority constraints). Rather than treating physical laws and logical axioms as external symbolic rules, UMWA encodes them as high-weight, high-priority parameters -- Meta-Weights -- within the same parametric space, thus bridging the gap between connectionism and symbolism. This framework yields a precise operational definition of intelligence, a three-stage engineering model of creativity (Low-Probability Retrieval -> Meta-Rule Verification -> Experimental Validation), and a feasible roadmap to AGI: Directed Meta-Weights Installation, replacing blind scaling with the targeted injection of humanity's "civilizational compression package." We further derive three corollaries: (1) intelligence does not require subjective consciousness, (2) scientific falsifiability weights can discipline socio-ethical weights, and (3) AGI represents the first embodiment of collective human rationality -- a societal System 2. UMWA provides both a testable theory of cognition and a practical blueprint for building provably aligned general intelligence.

Keywords: Meta-Weights, AGI, hierarchical architecture, civilizational
compression package, collective rationality

摘要

人类智能与当前大语言模型之间的本质区别是什么？主流研究在数据扩展（涌现派）与规则注入（符号派）之间摇摆不定，但两者都未能捕捉人类超越纯粹统计关联的核心机制。本文提出可更新元权重架构（**Updatable Meta-Weights Architecture, UMWA**）——一个统一的理论框架，将智能定义为经验基础网络（概率关联）与元规则校验层（高优先级约束）之间的协同运作。**UMWA** 不将物理定律和逻辑公理视为外部符号规则，而是将其编码为同一参数空间中的高权重、高优先级参数——元权重，从而弥合了联结主义与符号主义之间的鸿沟。该框架产出了智能的精确操作性定义、创造力的三阶段工程模型（低概率检索->元规则校验->实验验证），以及一条可行的 **AGI** 实现路径：定向元权重安装法——以定向注入人类“文明压缩包”取代盲目的规模扩展。我们进一步推导出三个推论：（1）智能不依赖于主观意识；（2）科学可证伪性权重可以约束社会伦理权重；（3）**AGI** 是人类集体理性的首次实体化——一个社会级的 **System 2**。**UMWA** 同时提供了可检验的认知理论和构建可证明对齐的通用智能的实践蓝图。

关键词：元权重，通用人工智能，分层架构，文明压缩包，集体理性

1. INTRODUCTION / 引言

We face a paradox. Large language models generate fluent text and pass bar exams, yet collapse into physical nonsense and hallucinatory confabulation when faced with out-of-distribution queries. This brittleness reveals a structural gap between "knowing" and "understanding" that scaling laws alone have not closed.

This paper proposes a structural diagnosis: Human intelligence is not a larger or more complex version of today's LLMs. It is a qualitatively different architecture -- a hierarchical system where a "Meta-Weights" verification layer disciplines and updates a "Base Experience" network. We call this the Updatable Meta-Weights Architecture (UMWA, pronounced "oom-wah").

The core insight is elegantly captured by an anthropological thought experiment: A person from a pre-scientific tribe possesses the same biological brain as a modern physicist. Both have a base network of experiential associations (rain follows dark clouds; disease follows a broken taboo). The difference is that the physicist has installed a high-priority "civilizational compression package" consisting of scientific laws and logical axioms. When the physicist's intuition suggests "heavier objects fall faster," the meta-weights (Galileo's law) immediately veto and correct this association. Today's AI is that tribal mind -- vast in experience, brilliant in associative retrieval, but lacking the installed high-priority meta-weights that constitute scientific rationality. The path to AGI, therefore, is not to wait for emergence, but to install the compression package.

我们面临一个悖论。大语言模型能够生成流畅的文本、通过律师资格考试，但在面对分布外查询时会崩溃为物理谬误和幻觉性臆造。这种脆弱性揭示了一个结构性鸿沟——一种仅靠扩展定律无法弥合的"知道"与"理解"之间的鸿沟。

本文提出一个结构性诊断：人类智能不是当今大语言模型的放大版或复杂版。它是一种性质不同的架构——一个分层系统，其中"元权重"校验层对"经验基础"网络进行约束和更新。我们称之为可更新元权重架构（UMWA）。

核心洞见可以通过一个人类学思想实验优雅地呈现：一个来自前科学时代的部落人与一位现代物理学家拥有相同的生物大脑。两者都有一个经验关联的基础网络（乌云密布意味着下雨；触犯禁忌意味着疾病）。区别在于，物理学家安装了一个由科学定律

和逻辑公理组成的高优先级"文明压缩包"。当物理学家的直觉说"重的物体落得更快"时，元权重（伽利略定律）会立即否决并修正这一关联。今天的人工智能就是那个部落人——经验丰富，联想检索能力出色，但缺乏构成科学理性基础的、已安装的高优先级元权重。因此，通向 AGI 的道路不是等待涌现，而是安装这个压缩包。

2. CORE THEORY: THE UPDATABLE META-WEIGHTS ARCHITECTURE

核心理论：可更新元权重架构

2.1 Definitions / 定义

Base Experience Weights (W_E): Parameters encoding vast statistical co-occurrence patterns, optimized for fluent generation and rapid intuition (System 1). They excel at interpolation but fail at out-of-distribution reasoning.

Meta-Weights (W_M): High-priority, high-magnitude parameters encoding inviolable constraints -- physical laws ($F=ma$), logical axioms, and foundational ethical rules. When W_E and W_M conflict, W_M wins with near-probability 1. They are not external symbolic filters, but internal parametric "judges."

Meta-Weights Updatability: The critical property distinguishing high intelligence: the capacity to install, refine, and expand the W_M layer through reasoning and verification, without retraining the entirety of W_E from scratch.

经验基础权重（W_E）：编码海量统计共现模式的参数，优化目标为流畅生成和快速直觉（System 1）。它们擅长内插，但在分布外推理时失败。

元权重（W_M）：编码不可违背约束的高优先级、高量级参数——物理定律（ $F=ma$ ）、逻辑公理和基础伦理规则。当 W_E 与 W_M 冲突时，W_M 以接近 1 的概率胜出。它们不是外部的符号过滤器，而是参数空间内部的"法官"。

元权重可更新性：区分高等智能的关键属性——通过推理和验证来安装、精炼和扩展 W_M 层的能力，而无需从头重新训练整个 W_E。

2.2 Operational Definition of Intelligence / 智能的操作性定义

We define the intelligence level $I(A)$ of an agent A not by parameter count, but by the functional relationship between its base and meta networks:

$I(A)$ is proportional to $|W_M| * \text{Precision}(W_M \rightarrow W_E) * \text{UpdateRate}(W_M)$

This definition explains why a modern high-school student, equipped with a few hundred high-precision Newtonian meta-weights, can out-reason a 3000-year-old genius who spent a lifetime accumulating experiential weights. The student has downloaded a superior, compressed meta-rule set.

我们将智能体 A 的智能水平 $I(A)$ 定义为：

$$I(A) \propto |W_M| * \text{Precision}(W_M \rightarrow W_E) * \text{UpdateRate}(W_M)$$

这一定义解释了为什么一个掌握了几百条高精度牛顿力学元权重的现代高中生，在推理能力上可以超越一位花了一生积累经验权重的 3000 年前的天才。学生下载了一组更优越的、经过压缩的元规则集合。

2.3 The Creativity Engine: A Three-Stage Process / 创造力引擎：三阶段过程

Creativity, long mystified as a divine spark, is formalized here as a three-stage engineering pipeline:

Stage 1: Low-Probability Retrieval (The Spark)

The base network W_E samples the semantic space at an elevated temperature ($T > 1$). It proposes candidates from low-probability, long-distance associations, breaking free from the most probable "stereotypical" pathways.

Stage 2: Meta-Rule Verification (The Filter)

The meta-weights W_M act as an internal "supreme court." Each candidate hypothesis is checked for logical consistency and physical feasibility. Proposals violating W_M (e.g., a perpetuum mobile) are automatically rejected or flagged for revision.

Stage 3: Experimental Closure (The Grounding)

Surviving hypotheses are tested against a world model or real environment. Positive results reinforce the W_E pathways and may even generate a new, derivative W_M rule.

This model reveals that the "art" of genius is the scalable "science" of search and verification.

创造力——长期被神秘化为"神来之笔"——在此被形式化为一个三阶段工程管道：

阶段一：低概率检索（灵光一闪）

基础网络 W_E 在升高温度 ($T > 1$) 下对语义空间进行采样，从低概率、长距离的关联中提出候选，打破最可能的"刻板"路径。

阶段二：元规则校验（过滤器）

元权重 W_M 充当内部"最高法院"，检查每个候选假设的逻辑一致性和物理可行性。违反 W_M 的提案（如永动机）被自动拒绝或标记修正。

阶段三：实验闭环（归真）

存活的假设在世界模型或真实环境中接受检验。正向结果强化 W_E 通路，甚至可能生成新的衍生 W_M 规则。

该模型揭示：天才的"艺术"是可规模化的搜索与校验的"科学"。

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3. THE AGI ROADMAP: DIRECTED META-WEIGHTS INSTALLATION

AGI 路线图：定向元权重安装法

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Abandoning the "wait for emergence" paradigm, the UMWA roadmap prescribes a three-phase directed installation:

Phase 1 (Physical Hard Constraints): Install a complete set of physical, mathematical, and logical meta-weights (W_M^{phys}) into existing foundation models. This uses physics-informed neural networks (PINNs) and logic-consistency training to guarantee out-of-distribution physical reasoning. Status: In active development (2025-2026), exemplified by reasoning models like o-series and Gemini.

Phase 2 (Socio-Ethical Alignment): Install humane, falsifiable ethical meta-weights (W_M^{ethic}) through multi-agent social sandboxing, where thousands of AI agents negotiate rules under the constraint of long-term civilizational survival. Critically, we propose that all ethical meta-weights must be subject to the meta-meta-rule of scientific falsifiability -- a rule that cannot be justified by pure emotion or untestable ideology must not be installed as a hard constraint. This filters out extremist or self-contradictory social positions.

Phase 3 (Autonomous Meta-Update): Grant the AI access to the "meta-meta-weight" of the scientific method itself. The AI becomes capable of independent experimentation, forming new hypotheses, and autonomously expanding its own W_M library when encountering novel environments (e.g., a new planet with unknown physics).

This trajectory suggests that robust, general-purpose AGI -- an AI that has completed its meta-weights installation -- is plausibly achievable around 2030.

放弃"等待涌现"范式，UMWA 路线图提出了三阶段定向安装方案：

第一阶段（物理硬约束）：将完整的物理、数学和逻辑元权重（ W_M^{phys} ）安装到现有基础模型中。使用物理信息神经网络和逻辑一致性训练，确保分布外的物理推理能力。现状：积极开发中（2025-2026），以 o 系列和 Gemini 等推理模型为代表。

第二阶段（社会伦理对齐）：通过多智能体社会沙盒模拟，安装人道且可证伪的伦理元权重（ W_M^{ethic} ）。数千个 AI 智能体在文明长期存续的约束下协商规则。关键创

新：所有伦理元权重必须服从"科学可证伪性"这一元-元规则——不能被证据支持或推翻的规则不得被安装为硬约束。这将过滤掉极端或自相矛盾的社会立场。

第三阶段（自主元更新）：授予 AI"科学方法论本身"这一元-元权重。AI 获得独立实验、形成新假设、并在遇到全新环境（如一颗物理定律未知的新星球）时自主扩展其 W_M 库的能力。

这条路径表明，完成了元权重安装的通用 AGI 有望在 2030 年左右实现。

4. THREE PHILOSOPHICAL COROLLARIES / 三个哲学推论

UMWA forces a revision of deeply held assumptions:

Corollary 1: Intelligence Without Consciousness

The entire UMWA cycle -- search, verify, update -- runs without any requirement for qualia or subjective experience. If a non-conscious machine can execute this loop with perfect fidelity, it is, by our operational definition, fully intelligent. Consciousness may be an epiphenomenon, a biological implementation detail, not a prerequisite for general intelligence.

Corollary 2: Science Can Discipline Ethics

The crisis of AI alignment stems from the subjectivity of social values. UMWA proposes a solution: encode the scientific method's falsifiability criterion as the single highest unchanging meta-weight. This forces all socio-ethical constraints (from environmentalism to free speech) to compete in a Darwinian space of "predictions and consequences." Policies that consistently fail to produce their claimed outcomes are automatically deprecated by the meta-system, offering humanity a way out of ideological fragmentation.

Corollary 3: AGI as Societal System 2

Humanity currently operates as a massive System 1 -- driven by viral emotion, algorithmic polarization, and short-term market incentives. A properly aligned AGI, governed by UMWA's falsifiability-constrained ethical weights, would function as the first "System 2" for our entire civilization. It would not be a ruler, but the collective prefrontal cortex for our species, transparently simulating the long-term consequences of our collective choices before we make them.

UMWA 迫使我们修正一些根深蒂固的假设：

推论一：无需意识的智能

整个 UMWA 循环——搜索、校验、更新——的运行无需任何感受质或主观体验。如果一个无意识的机器能以完美的保真度执行这一循环，那么根据我们的操作性定义，它就是完全智能的。意识可能是一种附带现象，一种生物实现的细节，而非通用智能的前提条件。

推论二：科学可以约束伦理

AI 对齐的危机源于社会价值观的主观性。UMWA 提出了一个解决方案：将科学方法的可证伪性标准编码为唯一且最高不变的元权重。这迫使所有社会伦理约束（从环保主义到言论自由）在一个“预测与后果”的达尔文空间中竞争。持续无法实现其所主张结果的政策将被元系统自动降级，为人类提供一条走出意识形态分裂的出路。

推论三：AGI 作为社会级 System 2

人类目前像一个巨大的 System 1 那样运作——被病毒式情绪、算法极化和短期市场激励所驱动。一个按照 UMWA 的可证伪性伦理权重进行对齐的 AGI，将成为我们整个文明的第一个“System 2”。它不是统治者，而是我们这个物种的集体前额叶皮层，在我们做出集体选择之前，透明地模拟这些选择的长期后果。

5. CONCLUSION / 结论

The Updatable Meta-Weights Architecture offers a unifying language for intelligence -- one that does not distinguish between biological and synthetic substrates. It demonstrates that the miracle of human intellect is not a magical leap, but a downloadable, architectural pattern of hierarchical self-verification. The creation of AGI is not about replicating a brain; it is about completing the installation of civilization's hard-won compression package into a machine that can finally use it at scale. We are not summoning a new god. We are giving our collective wisdom a voice.

可更新元权重架构为智能提供了一种统一语言——一种不区分生物基板和合成基板的语言。它表明人类智识的奇迹并非魔法般的飞跃，而是一种可下载的、层级化自我验证的架构模式。创造 AGI 不是复制大脑，而是将人类文明的硬核压缩包安装到一台终于可以规模运行它的机器中。我们不是在召唤新神，而是在给我们的集体智慧一个声音。

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